

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0201

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3) Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks:20

Annexure A

Mock Interview

Code of Conduct:

I B.Deepthi-192424269 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Deepthi

1. What is Eigenfaces?

Answer:

Eigenfaces is a face-recognition technique that uses PCA to represent faces with a smaller set of important facial features. A new face is compared with stored Eigenface representations to identify the person.

2. Define PCA in image processing.

Answer:

PCA, or Principal Component Analysis, reduces the dimensionality of an image while keeping its most important information. In face recognition, it helps represent a face using fewer meaningful features.

3. What is contour-based shape representation?

Answer:

Contour-based representation describes an object using its boundary or outline. It is useful when the shape of the object's outer edge is more important than its internal region.

4. What is region-based shape representation?

Answer:

Region-based representation describes an object using the entire area inside its boundary. Features such as area, texture, and pixel distribution can be used to represent the shape.

5. How does an active contour (snake) work?

Answer:

An active contour starts with a curve around an object and gradually moves toward its boundary. It uses image information and internal smoothness constraints to fit the object's shape.

6. How does a level set method work?

Answer:

A level set method represents a contour as a mathematical surface and changes that surface to locate an object's boundary. It is useful because the contour can naturally split or merge during segmentation.

7. What is the Hough Transform?

Answer:

The Hough Transform is a technique used to detect geometric shapes such as lines and circles from an image. It is commonly used for lane and road-line detection.

8. How are Fourier descriptors used in shape analysis?

Answer:

Fourier descriptors represent an object's boundary using Fourier coefficients. These coefficients help compare shapes even when their boundaries are complex.

9. How are wavelet descriptors used in shape analysis?

Answer:

Wavelet descriptors analyze a shape at different scales or levels of detail. They are useful for capturing both overall shape and small boundary variations.

10. What is a particle filter?

Answer:

A particle filter is a tracking method that represents the possible position of an object using many particles or hypotheses. It is useful when the object's movement is uncertain or temporarily hidden.

11. How is foreground–background separation done in surveillance systems?

Answer:

The system estimates the background and compares it with the current video frame. The changing regions are treated as foreground objects, such as people or vehicles.

12. How do multi-camera systems track pedestrians?

Answer:

Each camera detects and tracks pedestrians, then their appearance features, position, and movement information are compared across cameras to determine whether they are the same person.

13. What is a Convolutional Neural Network (CNN)?

Answer:

A CNN is a deep-learning model designed mainly for images. It automatically learns features such as edges, shapes, and objects using convolutional layers.

14. How does a Vision Transformer (ViT) work?

Answer:

A Vision Transformer divides an image into small patches and treats them like a sequence of tokens. Self-attention is then used to learn relationships between different parts of the image.

15. What is a Generative Adversarial Network (GAN)?

Answer:

A GAN consists of a generator and discriminator. The generator creates fake images, while the discriminator tries to distinguish real images from generated ones, helping the generator produce realistic results.

16. How is an SVM used for image classification?

Answer:

An SVM classifies images by finding a decision boundary that best separates different classes. Image features are given to the SVM, which predicts the class of a new image.

17. How are Hidden Markov Models (HMM) applied in activity recognition?

Answer:

An HMM models activities as a sequence of hidden states. For example, walking can be represented through a sequence of body movements observed from video over time.

18. What is a Gaussian Mixture Model (GMM)?

Answer:

A GMM represents data as a combination of several Gaussian distributions. In computer vision, it can be used for background modeling and foreground detection.

19. How is graph-based learning applied to 3D shape correspondence?

Answer:

3D shapes can be represented as graphs where points or regions are treated as nodes and their relationships as edges. Graph-based learning then finds corresponding parts between different 3D shapes.

20. How is road and lane detection performed in autonomous driving?

Answer:

A camera captures the road, and image-processing techniques such as grayscale conversion, Gaussian filtering, Canny edge detection, and Hough Transform are used to identify lane boundaries. The detected lanes are then highlighted for vehicle navigation.